

Emergence and Artificial Life

“Anyone who looks at living organisms knows perfectly well that they can produce other organisms like themselves. . . . Furthermore, it’s equally evident that what goes on is actually one degree better than self-reproduction, for organisms appear to have gotten more elaborate in the course of time. Today’s organisms are phylogenetically descended from others which were vastly simpler than they are, so much simpler, in fact, that it’s inconceivable how any kind of description of the later, complex organism could have existed in the earlier one.”

—John von Neumann⁶³

THE PROBLEM OF EMERGENCE

The problem of emergence classically involved the origins of qualitatively new structures and functions which were not reducible to those already in existence,^{5,20,32,33,46,64} In its most general form it encompasses all questions of the appearance of fundamental novelty in the world, the origins of order, and the increase in hierarchical complexity.^{25,39,40,47,52,59}

The pragmatic relevance of emergence is intimately related to Descartes Dictum: how can a designer build a device which outperforms the designer's specifications?² If our devices follow our specifications too closely, they will fail to improve on those specifications. If, on the other hand, they are not in any way constrained by our purposes, they may cease to be of any use to us at all. Thus, the problem of emergence is the problem of specification vs. creativity, of closure and replicability vs. open-endedness and surprise. Its solution entails finding ways of building devices which circumvent Descartes Dictum. Emergent devices to be useful in amplifying our own creativity must have both a degree of structural autonomy relative to us as well as richness of potential structure. There has been a long-standing debate—from Leibnitz to Lady Lovelace to the present—over whether purely computational devices are capable of fundamentally-creative, truly emergent behavior. This paper will discuss various kinds of devices capable of emergent behaviors and take up the question of whether we can by purely computational means amplify our capacities as observers and actors in the physical world.

EMERGENCE: COMPUTATIONAL, THERMODYNAMIC, AND RELATIVE TO A MODEL

Currently there are three largely separate discourses on the problem of emergence, each with its own interpretation of the problem and distinct view of the world (Table 1). These three major conceptions of emergence might be called *computational emergence*, *thermodynamic emergence*, and *emergence relative to a model*.

Computational emergence is the view that emergent, complex global forms can arise from local computational interactions.^{15,21,22} The result is a “bottom up” computational approach which is highly compatible with connectionist ideas now in vogue. Because its ontology only admits of micro-deterministic computational interactions (of which chaotic processes can generate apparently nondeterministic macro-interactions), there is a strong platonic component to the world view: the ideal forms of computational behaviors can be abstracted completely from their material substrates, and the material world can be left completely for a virtual one. In assuming rule-governed, bottom-up organization rather than semi-autonomous levels of organization, computational emergence tacitly incorporates the older reductionist assumption that micro-orders determine macro-orders but not vice versa.²⁰ There is a small literature which has critically addressed some of the specific problems with generating emergent behaviors in computer simulations in particular and micro-deterministic processes in general.^{7,8,12,23,37,39,40,44,45,52,55,56,57,58} While computational emergence has been slow to provide adequate, nonarbitrary definitions of exactly what distinguishes emergent computations from nonemergent ones, some efforts are currently being made to sharpen the criteria.¹⁴ *If we randomly come across a computer simulation and we have no clue as to its purpose, can we tell if its computations are emergent?*

TABLE 1 A survey of three contemporary discourses on emergence

theory	account of the origins of order	ontology	exemplars	research program
Computational Emergence (formally based)	order-from-order	Discrete Universe	phase changes	realize emergent behaviors through cellular automata and evolutionary computer simulations. (Langton, Toffoli)
	Macro-order from micro-determinism	Monism	simulated turbulence	
	Macro-indeterminism through mathematical chaos	Microscopic Computational Rules	Kauffman nets	
		Platonist	chaos & fractals in nature	
Thermodynamic Emergence (physically based)	Order-from-noise	Continuous Universe	origins of life	Develop a thermodynamic theory to describe how stable, complex structures can arise far from equilibrium. Apply this to biological & social systems & upwards. (Prigogine, Iserall)
	discrete macro-structures (symbols=attractors) from continuous micro-processes	Monism	condensation of DNA code	
	New structures emerge through fluctuations	Continuous Physical Laws	catalytic networks	
		Materialist	formation of dissipative structures	
Emergence Relative to a Model (functionally based)	Form-from-formlessness	Symbol-Matter	evolution of new	Realize emergent functions through the construction of semantically adaptive devices. Augment the capabilities of the observer. (Pattee, Pask, Cariani)
	Order-from-chaos	Hylomorphism	sensory & effector organs	
	Definition-from-ambiguity	Complementarity between Different Points of View	construction of new tools, languages, & models	
	Processes of linking symbols to the world result in new functions and extend the realm of symbolic activity	Pragmatist		

Thermodynamic emergence finds its way into the Artificial Life discourse through realizations (osmotic growths), simulations (reaction-diffusion-systems, autocatalytic cycles), and dynamical theory (attractors and dissipative substrates of symbolic activity), although the necessary thermodynamic underpinnings of a theory of life have yet to be considered by the community at large. Active efforts are underway to connect the emergence of thermodynamic structures with theories of living organization,^{12,48,60,65} perspectives which should be highly relevant to Artificial Life as a whole in the future. A major question is how to connect thermodynamic theories of structural stability with the appearance of new *functions*. Given a thermodynamic description of a material system, can we deduce which (kinds of) functions can or cannot be implemented?

The emergence-relative-to-a-model perspective arises out of the epistemological issues surrounding the successive enlargement of perceptual, cognitive, and behavioral organismic repertoires over biological evolution and their counterparts in the evolution of scientific models. These issues necessarily involve the evolutionary origins of symbols and coding relations in biological systems as well as the appearance of novel functions over time which are not reducible to previously existing functions. Hence while computational emergence is concerned with what kinds of global behaviors can be built up through the operation of local rules without altering the local rules, emergence relative to a model is concerned with the formation of global structures which then constrain and alter local interactions.^{37,38,39,40} An example is the formation of new sensory organs, whose global interaction patterns vis-à-vis the world lead to metabolic changes contingent upon the sensed state of the world. While computational emergence sees macroscopic order as a consequence of microscopic order, theoretical biology sees the symbols and coding relations as arising through nonsymbolic, rate-dependent dynamics. Thus, it is always important to bear in mind that the discrete symbols in biological systems have nonsymbolic material substrates, and the origins of new symbolic primitives must be organized out of this nonsymbolic, analog realm.

While computational emergence is primarily a formal, mathematically based conception, and thermodynamic emergence is a physically based materialism, emergence relative to a model can be seen as a functionally based hylomorphism.²⁹ Here matter has as many properties as one can measure; a given object can support radically different types of behavior, depending upon how one has chosen to observe it. In addition, many of the functions and organizational properties of interest cannot be reduced to purely structural descriptions; we are confronted directly with the problem of defining and recognizing living organization and biological functionalities in the structure-oriented realm of our simulations.^{17,16,37,39,40,51,52,53} How do we come to recognize the "amoeba in the pot" if we are only given the local concentrations of the chemical species? Clearly, the concept of an amoeba is not reducible to a set of points in a multidimensional chemical concentration space. While logical and structural descriptions can generally be ordered into consistent formal relations, functional measures usually confound complete translation from other measures (e.g., swimming reduced to floating and kicking and not drowning and...). To explain the origins of new functionalities, in our models as well as in the

biological organisms themselves, one needs new observables to appear which are not simply logical combinations of pre-existing ones. This is the central epistemological problem of creating new observational primitives, and the key to constructing devices with the means to create their own functional categories.³⁶ If we simply specify the observational primitives outright, as we do for the primitive features of a connectionist device, then the device will not have the structural autonomy needed to construct new primitives, and its usefulness will be limited to those tasks it can perform within the categories we have given it. To build devices which find new observational primitives for us, they must be made epistemically autonomous relative to us, capable of searching realms for which we have no inkling. But to embark on this very ambitious enterprise, we must be first be very clear exactly how we would recognize the emergence of a new observational primitive in an organism or device we are observing. We need a precise definition of emergence.

EMERGENCE RELATIVE TO A MODEL

The emergence-relative-to-a-model view^{8,54,55,56} sees emergence as the deviation of the behavior of a physical system from an observer's model of it. Emergence then involves a change in the relationship between the observer and the physical system under observation. If we are observing a device which changes its internal structure and consequently its behavior, we as observers will need to change our model to "track" the device's behavior in order to successfully continue to predict its actions. This perspective is close to the conception of "self-organizing systems" proposed by Ashby, Pask, Beer, and von Foerster thirty years ago.^{4,36} While Howard Pattee and Robert Rosen have explicated similar concepts using the language of dynamical systems, this paper will present a complementary account in terms of systems-theoretic distinctions and cybernetic mechanisms.

THE MODELLING RELATION

The basic modelling methodology utilized here was first explicated by von Helmholtz and Hertz a century ago, and elaborated by the physicists Bohr and Heisenberg as they dealt with the foundational problems raised by quantum mechanics (Figure 1). The framework was further extended by Ashby³ to form the basis of systems theory and by Rosen^{51,54,55,56} to form the basis for a theoretical biology.^{16,18,19}

In this perspective the modelling framework consists of two kinds of processes: measurements, which relate the symbols in the formal part of the model to the world at large, and computations, which relate the symbols in the formal part to other symbols via rule-governed operations. In semiotic terms, computations are logically

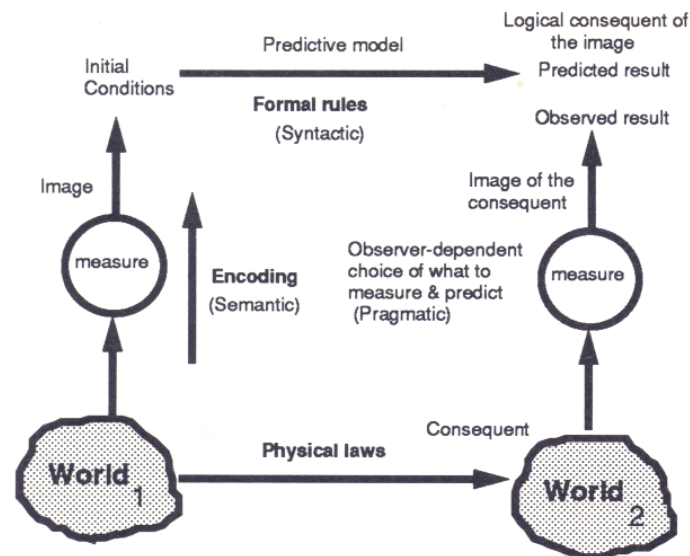


Figure 4.1. The Hertzian modelling paradigm

FIGURE 1 Semiotic relations in scientific models. The Hertzian commutation relation between symbols and the world.

necessary, convention-governed *syntactic* operations, while measurements are empirically contingent, materially governed *semantic* operations. The choice of appropriate measurements and computations which determine how well the model satisfies the purposes of the observer constitute intentional, *pragmatic* operations. These three kinds of operations, syntactic, semantic, and pragmatic (necessary, contingent, and intentional, respectively), form the irreducible orthogonal functional axes of the modelling relation. Thus a model consists of (1) a set of observables, implemented through contingent connections to the world (measuring devices, sensors), (2) a set of formal relations on the states of the observables implemented through physical devices with reliable state-transition behavior, (3) a set of state variables of interest which the observer seeks to predict, (4) evaluative criteria for that framework, and (5) means of changing either the measurements or the computations made (by building or recalibrating measuring devices or by changing computational rules, reprogramming). All of these aspects of the modeling relation can be made accessible to a public realm of a community of observers (Helmholtz's *locus observandi*) through calibration processes. The much-discussed tacit parts of scientific theories, the images and conceptual contexts through which the observers interpret these relations, affect the choices of observables and computations, but not their results once they have been selected.

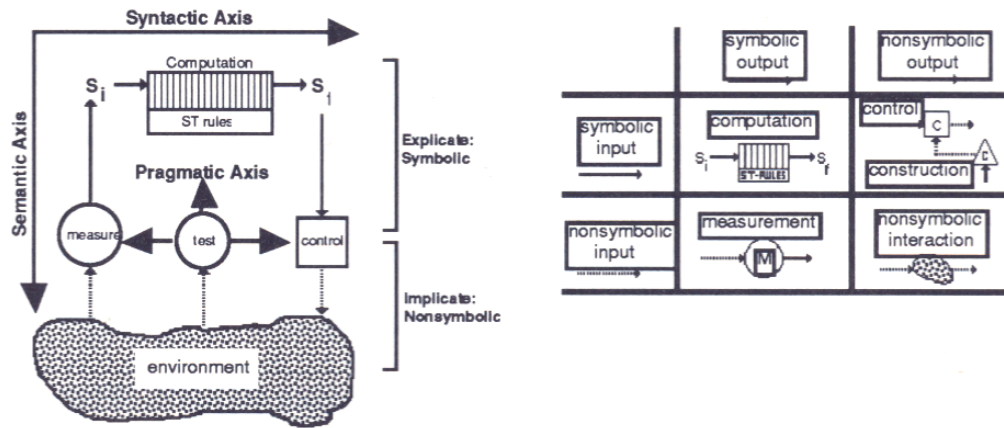


FIGURE 2 (a) Basic semiotic relations in organisms and devices. (b) Basic semiotic operations.

This basic modeling framework can be enlarged to describe devices and organisms by adding a set of effectors through which the device or organism influences the world beyond its boundaries.^{8,24,39,40,55,56,58} *Thus the epistemological framework used to describe the operations of the modeling relations can also be used to describe modeling relations as they appear in organisms, devices, and other observers.* If we consider a robot with sensing elements, computational elements, and effector elements, then the corresponding basic operations will be those of *measurement*, *computation*, and *control*. Since effectors take the symbolic outputs of the computational part and act on the nonsymbolic world, they, along with measurements, determine the relation of the symbols in the computational part of the device to the world at large. Hence both measurements and controls are semantic operations.

Because a physical system will always contain multitudes of variables which by necessity will lie unrepresented in our finite models and because the choice of the subset of the relations that we choose to include in our models is up to the observer, it is imperative that we specify how we are viewing the system (what are the observables that constitute our *observational frame*) when we construct our definitions or make statements about the physical world. Depending upon how we look at the electronic devices on our desks (e.g., even in terms of volts, millivolts, or microvolt descriptions), we can see the material system as either completely rule governed or highly stochastic. By specifying the observational frame, we can avoid unproductive arguments over whether the device is “really” deterministic or nondeterministic—we can subject the question to an unambiguous empirical test and come to an agreement, on the outcome.

OBSERVED SYSTEMS AND THEIR BEHAVIORS

It is helpful to adopt Ashby's concepts of "system" and "state" in these discussions. A *state* is a joint property of an observer and the physical system being observed, an observable distinction: "By a state of a system is meant any well-defined condition or property that will be recognized if it occurs again."³ A *system* consists of a set of distinguishable states chosen by the observer. The behavior of this system will consist of the patterns of state transitions that it undergoes over some period of observation (Figure 3). These state transitions can be apparently determinate (e.g., $4 \rightarrow 5$) or they can be apparently indeterminate (e.g., $16 \rightarrow 4$, $16 \rightarrow 18$), depending upon whether a given predecessor state is always observed to transit to the same successor state. Within this frame, those transitions which are determinate can be described in terms of a rule relating the states (4 and 5); such that one can duplicate exactly the apparent behavior of the system by using this rule. When a such rule can be written, the state can be treated as a symbolic type, abstracted from its material substrate. In this situation we lose no information about the trajectory of the system if we drop the distinction between the two states (4 and 5) by merging them. We can label such state-determined "apparently necessary" transitions as computations and nondeterministic, contingent transitions as measurements, controls, and nonsymbolic transductions, depending upon how they are connected to computational transitions.⁸ The method gives us a clear means of deciding whether we are seeing a computation, measurement, control, or nonsymbolic interaction taking place in a material system under observation. Immediately we can decide whether a given material system observed via a particular frame is implementing "computations." This a distinct advantage over purely formal definitions of computation, whose mapping to the physical world is left unspecified (e.g., Is the amoeba in our test tube a Turing machine? On what basis do we decide?). Likewise, these criteria offer a possible solution to "the measurement problem," the question of when a measurement is judged to have taken place.^{44,45}

Those states that participate in state-determined, computational transitions can be considered "symbolic" states, and the rest can be considered "nonsymbolic" states. To simplify the discussion, we will talk in terms of computational, syntactic transitions (those which can be described in terms of rule-governed transitions between symbolic states) and noncomputational, semantic transitions (those contingent measurement and control transitions which connect symbolic and nonsymbolic states). Those transitions in which no symbolic states participate can be identified as nonsymbolic interactions, and these transitions are neither syntactic nor semantic, since no symbolic states are involved. In effect we have systems-theoretic, operational definitions for the basic operations of the observer-actor, be it a device, organism, or some other appropriately organized physical system. This also gives us the means of distinguishing between syntactic, semantic, and pragmatic relations.

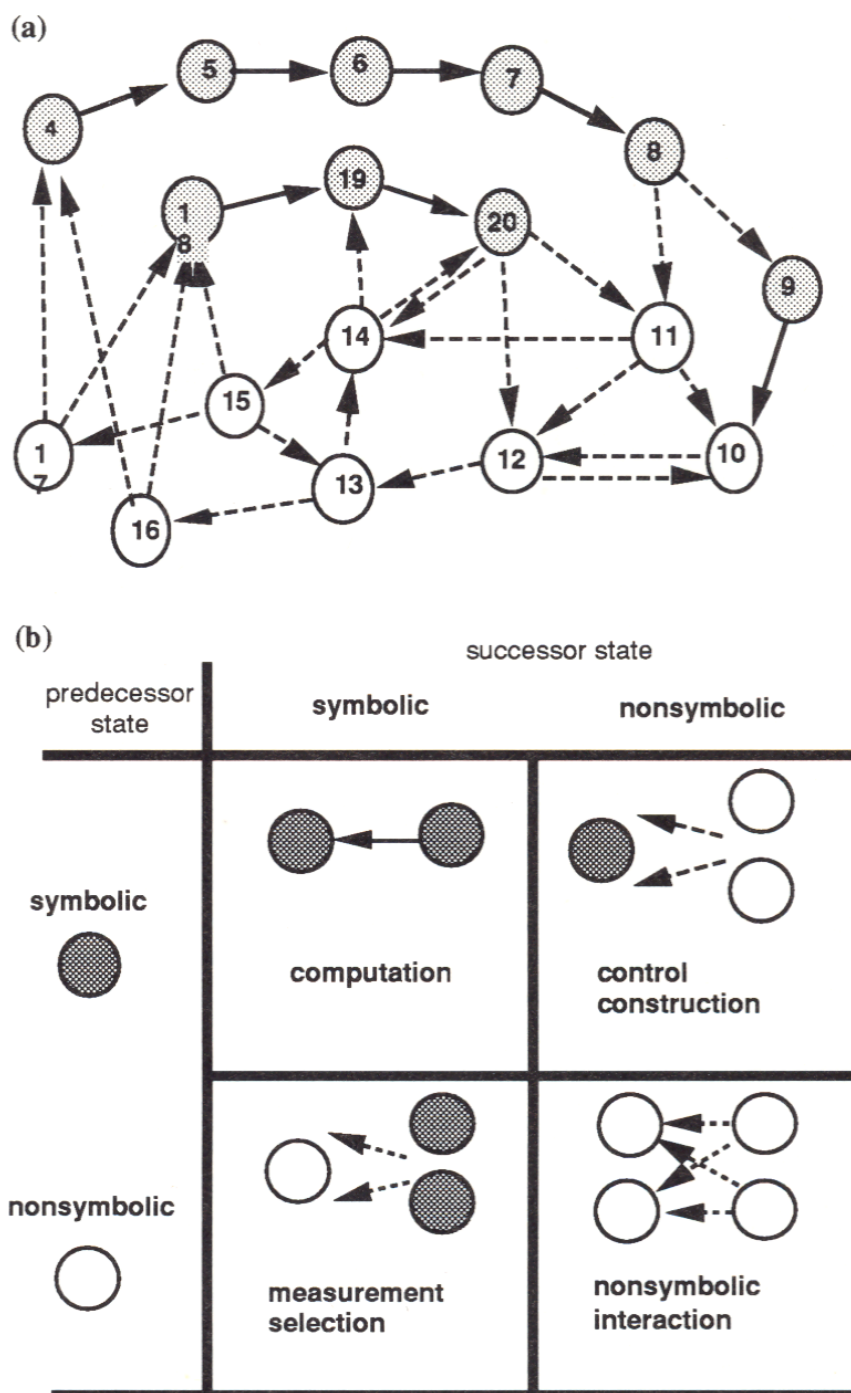


FIGURE 3 (a) An observed state diagram. (b) Observed-state correlates of the basic semiotic operations.

The result is a phenomenally grounded *semiotic functionalism* incorporating syntactic, semantic, and pragmatic relations, rather than a logically grounded computational functionalism which only considers syntactic operations as its constituent relations.

These distinctions are also relevant for an operational definition of living organization, a central theoretical problem for Artificial Life. If we lack such an empirically decidable definition, we have no means of deciding whether our simulated organisms are living or not, and no means of evaluating our research efforts.^{13,30,31,37,52,53} Thus far, the most thoughtful attempts at a definition have come from theoretical biology and cybernetics and have as their basis a network which is constantly reproducing its parts through a recurrent set of production relations: Rashevsky's "relational biology,"⁵⁰ metabolism-repair (M, R) systems,^{28,55,56} semantically closed systems,^{37,42} autopoietic systems,^{13,26,30,31,62} self-modifying systems,^{11,16,18,19} self-reproducing automata,⁶³ and recurrent neural networks.²⁷ The semiotic functionalism outlined above may provide a means for grounding our recognition of the organism/environment boundary: regions of computations (rule-governed order) bounded by contingent measurement and control transitions.

TYPES OF EMERGENT BEHAVIOR

"The main point to realize is that all knowledge presents itself within a conceptual framework adapted to account for previous experience and that any such frame may prove too narrow to comprehend new experiences."

—Niels Bohr⁶

Once we have fixed our observational frame we can talk precisely about emergence: whether the behavior of the physical system in question has changed with respect to the frame and in what ways it has changed. If we observe the system for a period of time, we can build up a model including all of the syntactic and semantic state transitions we have previously observed and we can see if new state-transition patterns appear over time as the system subsequently transits through its states. The model thus constitutes the observer's expectations of how the system will behave in the future. If we are confident that our measuring apparatuses are stable (via calibration tests), then any observed deviations from the observer's model can be reasonably interpreted as structural changes in the organism or device under observation.

If no new state-transitions arise, then the device or organism is *nonemergent* with respect to the observer's frame. The observer would need to make no change in his/her model in order to include all the observed state transitions within the system. We would infer from this nonemergent behavior that the structure of the organism or device was stable relative to our observational frame. Whatever other

physical processes are going on in the material system, they do not affect the relation between the observer and the material system.

If the pattern of syntactic state transitions changes such that new computational transitions are formed, then the device or organism will appear to be *syntactically emergent*. From such behavior, we could infer that the device or organism was implementing a new computation within the same set of symbolic states. We could interpret this behavior as signifying the switching of the computations performed by the device or organism, contingent upon some other process or mechanism unrepresented in the model.

If the pattern of semantic state-transitions changes such that new measurement or control transitions are formed, then the device or organism will appear to be *semantically emergent*. This would signify changes in the structure of the device which make its semantic operations contingent upon other factors outside the observational frame. In the case of an emergent measurement, the organism or device would appear to have acquired the ability to sense changes in the world which were unrepresented in the observer's model. The observer could then have access to this new organism/device observable by watching the state-transition behavior of the organism/device. An observer trying to track the device would find it necessary to add an observable to the model whenever a new sensor was constructed by the device. The apparent dimensionality of the device's state space would thus increase by one.

Syntactic and semantic emergence thus are not mutually exclusive possibilities: both processes can be happening simultaneously in the same system.

EMERGENCE AND ADAPTIVE DEVICES

What sorts of devices exhibit the various types of emergent behavior? A comprehensive taxonomy of adaptive devices incorporating combinations of fixed and performance-dependent sensing, computing, and effecting elements can be developed corresponding to the types of emergent behaviors listed above.^{8,36,66} According to the plasticities of the respective functionalities, the capacities and limitations of each type of adaptivity can be outlined (Table 2). Nonemergent devices (Figure 4) are those with fixed syntactic parts (nonadaptive computational devices) and fixed semantic parts (nonadaptive robotic devices). Syntactically emergent devices (Figure 5(a)) are those devices having performance-contingent syntactic parts (adaptive devices). Semantically emergent devices (Figure 5(b)) are those with performance-contingent semantic parts (evolutionary devices). Devices which are both syntactically and semantically adaptive (general evolutionary devices, Figure 6(b)) would appear both syntactically and semantically emergent.

TABLE 2 Summary of capacities and limitations of some of the device types

device type	plasticity	capacities	limitations
fixed-computational	fixed syntax	reliable execution of pre-specified rules	limited to pre-specified rules and states
fixed-robotic	fixed syntax fixed semantics	reliable execution of fixed percept-action combinations	no feedback or learning from environment
adaptive	adaptive syntax fixed semantics	performance-dependent optimization of percept-action coordination	limited to precept & action categories fixed by the sensors & effectors
general evolutionary	adaptive syntax adaptive semantics	creation of new percept & action categories: performance-dependent optimization within these categories	time to construct & test new sensors & effectors may be very long

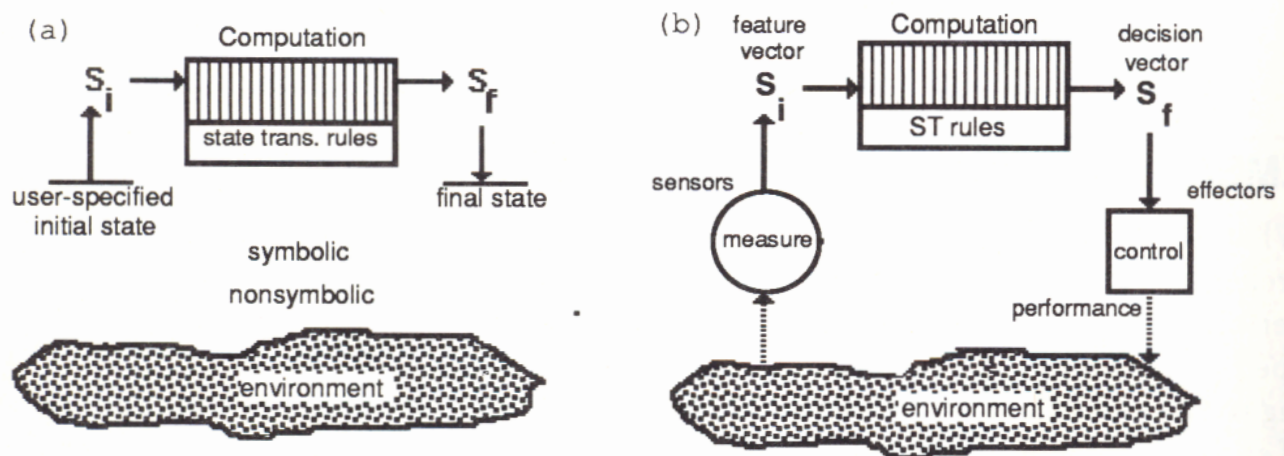


FIGURE 4 Two types of nonadaptive devices: (a) fixed computational and (b) fixed robotic devices.

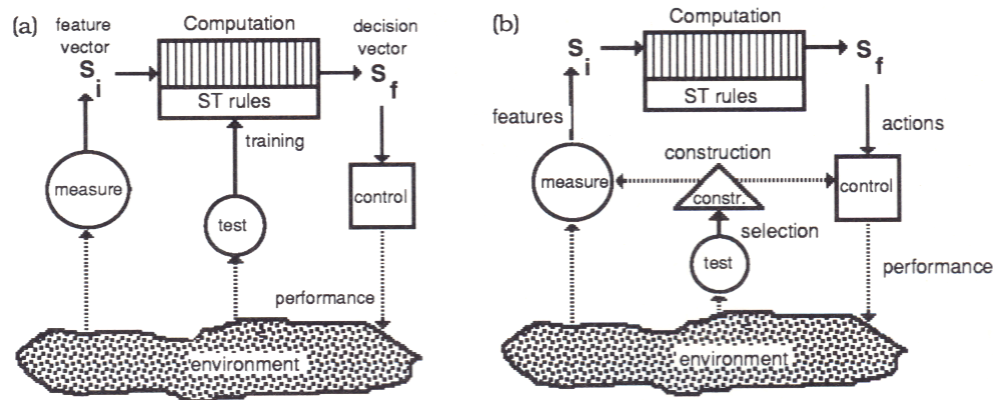


FIGURE 5 Two types of adaptive robotic devices: (a) syntactically adaptive and (b) semantically adaptive.

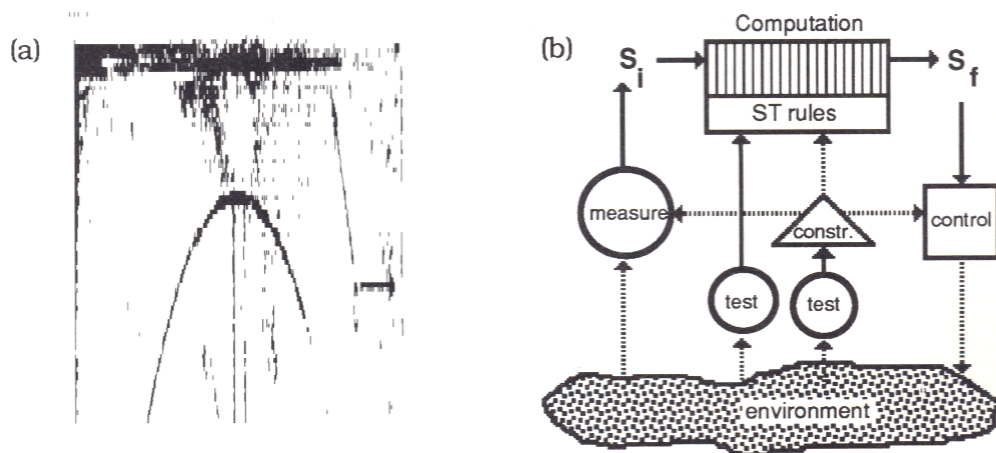


FIGURE 6 (a) Pask's electrochemical device (left) and (b) a general evolutionary device (right).

Nonadaptive computational devices. If we examine a computer with fixed programs, and we take as our observational frame the total machine states of the device, then the device will appear to us as a completely state-determined system.

Fig. 6a. Corrigenda. Image misrepresented due to production error. Correct image is at the end of the paper.

The device will yield consistent, replicable results—we can implement a formal system (in Hilbert's sense) using the states of the device as our tokens. As long as there are no hidden inputs upon which the computational outcome depends, then the device will continue to behave in the same way given the same initial states. No new state transitions will be created when we run the device, since it has been designed to perform independently of fluctuations in its environment. Hence the device will appear to be nonemergent relative to the observational frame of total machine states. There will be other physical processes going on within the device (heat flows, humidity changes, magnetic fields), but they will not make a difference on the behavior as seen through our observational frame of all its discrete-voltage gate states.

The same is true for computer programs and simulations, which have some subset of the total machine states as their states. As long as we are observing all of the program and simulation states (state-variable values, pseudo-random number generator states) and all the input values, then the program/simulation will give us the exact same behavior over time, and will appear nonemergent. We will discuss the implications for artificial life simulations below.

Nonadaptive robotic devices. Likewise fixed robotic devices will have nonemergent behavior, owing to the stability of their sensing, computing, and effecting elements. The sensing and control elements will have contingent transitions, but no new ones will arise if these elements are stable and enough time has been taken to exhaust the repertoire of possibilities.

Syntactically-adaptive devices (or adaptive computational devices) alter their computational parts contingent upon their performance in an external environment. Typically the device incorporates some performance measure and training rules which alter its computational part to improve performance. Consequently, if the observer chooses his/her frame to coincide with the computational states of the device, the pattern of computational transitions will change with every training iteration. Between training iterations the pattern of computational transitions will be stable and the device will appear nonemergent. Trainable classifiers and controllers, neural nets, and genetic algorithms will appear to be syntactically emergent as long as performance measure on which the computational changes are based appears as a contingent event to the observer. Typically this is the case. The user of an adaptive device does not know beforehand whether a given computational change will improve or degrade the device's performance, so the testing of the change appears contingent.

Semantically adaptive devices construct and select their sensing and effecting elements contingent upon their performance. The evolution of sensory and effector organs, the immune system, and activity-dependent, sensory-motor specializations (the musician's ear, the athlete's reflexes) are natural examples. The development of sensory prostheses and scientific measuring instruments (telescopes, radiometers, molecular assays, etc.) and effector prostheses (hand tools, motor vehicles, weaving looms, chemical plants, etc.) are artificial examples if their human designer-constructors are included.

Apparently there is but one device in the engineering literature with the explicit goal of adaptively creating new sensors *de novo*.^{34,35,36} In the late 1950's the cyberneticist Gordon Pask constructed an electrochemical device along the lines of an adaptive coherer (Figure 8(a)). Although a stable analog, connectionist network could have been implemented with his apparatus, Pask designed his device to be conditionally sensitive to nonelectrical perturbations in its environment, such that it could be tuned with the appropriate rewards. Given a set of electrodes in an aqueous ferrous sulfate/sulphuric acid solution, ferrous threads could be adaptively grown to become sensitive to sound. In about half a day the assemblage could be trained to discriminate between two frequencies of sound: "the evolution of an ear."³⁵

A device of this type could be combined with an adaptive computed neural network to form a general evolutionary device (Figure 8(b)): the electrochemical assemblage would evolve the feature primitives; hence the semantics of the feature space that the neural network would adaptively partition. We could thereby achieve adaptation on both syntactic and semantic levels: the resulting assemblage would have the means of both creating its own feature primitives and adapting its computations within those sets of primitives. Such a device would be *epistemically autonomous*, capable of choosing its own semantic categories as well as its syntactic operations on the alternatives.

A network of such devices would be capable of generating new, independent inter-element signalling channels in an apparently open-ended manner, thus increasing the dimensionality of its signal space. Such considerations may be relevant to multiplexing and creation of new signalling possibilities in real neurons.¹⁰ A still greater degree of autonomy would entail devices which not only determined their own syntactic and semantic relations, but their pragmatic relations as well. Such *motivationally autonomous* devices would have mechanisms for constructing their own performance-measuring apparatuses, hence the criteria which control the selection of their syntactic and semantic functions. Such devices would not be useful for accomplishing *our* purposes as their evaluatory criteria might well diverge from our own over time, but this is the situation we face with other autonomous human beings, with desires other than our own, and the dilemma faced by all human parents at some point during the development of their children.

IMPLICATIONS FOR THE INTERPRETATION OF ARTIFICIAL LIFE SIMULATIONS

The emergence-relative-to-a-model view has deep implications for the interpretation of artificial life simulations. All computer simulations can be described in terms of finite-state automata, as networks of computational state transitions, as formal symbol manipulation systems. As observer-programmers we can always find a frame which will make our simulation appear nonemergent. If we choose our observables

to coincide with the stable computational states of the finite state automaton being implemented by the simulation, then we will always see it as a nonemergent state-determined system. Here the states of our frame will generally correspond to the discrete values that the state variables of the simulation can take on and the state transitions will correspond to all of the simulation rules which govern the values of the state variables. Every time the simulation is run with the same initial conditions, the simulation will transit through the same trajectory of variable values. The computer simulation will be completely replicable; there will be no deviation of the simulation's behavior from the model of possible trajectories built up by the observer. Thus, from this perspective many of the breakout strategies that have been proposed to make artificial life simulations "open-ended" and "emergent" will simply not improve the situation because they do not change the formal, completely replicable nature of the process. Increasing the size of the simulation, adding new layers of simulation rules, simulating random or chaotic processes, or representing genotypes and phenotypes will not in any way change the replicability of the simulation; hence these changes will be ineffective at transforming a previously nonemergent simulation into an emergent one.

Likewise, the inclusion of global patterns, such as cycles, waves, gliders, or other complex forms into the consideration of cellular automata behavior also fails to change the replicability of these simulations. Typically, the appearance of these patterns is thought to be an emergent event in the device itself: previously the pattern was absent, now it is present. For the purposes of judging whether an emergent event has occurred, we need to be careful not to shift frames of reference in these situations, from talking in terms of microstates and pixel states before and "higher level" features afterwards. If we start to observe the device in terms of individual pixels, we must continue to do so in those terms throughout. If we wish to include complex pixel patterns (e.g., cycles, waves, moving patterns which look to us like a horse galloping), they need to be in our state descriptions from the start, or they will remain in the realm of tacit, private observation, unrecognized by our public model. These other distinctions can then added to the observational frame, but as long as the mechanisms for recognizing the complex pixel patterns are reliable and this recognition is solely a function of the pixel pattern presented, the replicability of the observed behavior of the simulation will be unaffected; hence the simulation will still appear nonemergent. That the behavior of the cellular automata by itself is nonemergent should not obscure its potentially emergent effects on the people intently watching the screen. They are busy recognizing new patterns and forming new concepts. *The interesting emergent events that involve artificial life simulations reside not in the simulations themselves, but in the ways that they change the way we think and interact with the world.* Rather than emergent devices in their own right, these computer simulations are catalysts for emergent processes in our minds; they help us create new ways of seeing the world.

MEASUREMENTS AND CONTROLS IN SIMULATED ORGANISMS

One of the hopes for artificial life is that we can gain useful knowledge of the material world around us by the construction of computer models and simulations. We know that various computational functions can be carried out by higher levels of computational organization—the subcognition as computation approach. It has been a matter of debate within the community, however, whether artificial life simulations are capable of realizing measurements, as opposed to merely representing them.^{44,45} At the outset, there should be no doubt in anyone's mind that we cannot make measurements on the material world outside of our computers solely by programming computations within our computers. *Operationally we cannot carry out measurements on the world about us if we completely control the actions of the measuring devices.* The measurements become a consequence of our actions rather than a process contingent upon factors outside our control. An example would be if we immersed a thermometer in a heat bath whose temperature we controlled completely—the readings we would get would cease to tell us anything about the world outside the heat bath. If we remove the human controller and close the loop by connecting the output of the thermometer to the heating element of the bath, then the thermometer-bath system will appear as a circular series of state-determined computations. To the extent that the system is not contingent upon fluctuations from outside the bath, it is informationally closed with respect to that external world.

Our simulated organisms are likewise part of informationally closed computational cycles, purposely insulated from the nonsymbolic world outside the simulations. This is exactly why we cannot make measurements on the physical world through programs and simulations that we specify completely. From the vantage point of total simulation states, simulated organisms appear as completely specified syntactic objects, having completely rule-governed behavior, and any contingent operations we have attempted to implement (e.g., perceptions) will lose their contingency once they are embedded in the simulation. Our simulated organisms are like the heat bath we control: they reflect our actions perfectly; hence we only learn about the world of artificial conventions we have constructed. *Whatever information or knowledge we gain from performing computational operations, it cannot be the same kind of knowledge or information we gain by making measurements.* We face the analytic-synthetic distinction in yet another context, the general question of exactly what it is we learn when we do mathematics.⁴⁹

Perhaps we cannot make measurements through our simulations, but what about our simulated organisms on *their* simulated environments? The important question here is not whether the organism or robot is “simulated” or “material,” since as limited observers we see the system only through a series of state transitions (we don't know what is on the other side of our measuring devices that give us our observable states). The critical question becomes how do we distinguish a measurement process in our simulation from all the computations? If the *form* of

the behavior is all that matters, we should be able to do this with an unlabelled, uninterpreted simulation. But if we have access to all of the simulation states, then our simulation will appear to us as a large state-determined system. There is no way to make the distinction, since all state transitions will appear to be necessary ones.

It has been suggested that a way out of this dilemma is to “take the perspective of the artificial organism” by discarding our knowledge of the simulated environment. As external observers, if we restrict our states to those of the organism and leave the simulated environment out of the system, then the organisms will indeed appear to be implementing contingent operations and there will be apparent informational exchange. We as external observers who no longer have access to the total simulation states can now gain empirical information about the simulated environment, since we no longer control it. Once we do this, however, we cannot vacillate between ignorance and omniscience. Once we have disavowed all knowledge of the environment, we can no longer interpret our simulated organisms as purely formal entities, since their internal states will be contingent on “hidden variables” unrepresented in our truncated frame. Either we view our simulated organisms through the lens of total simulation states and see them as purely computational, nonemergent formal objects, or we view them through a restricted lens and see them as informationally open, potentially emergent systems implementing measurements and controls as well as computations. *We cannot see both pictures at the same time.*

Finally, because all of our simulations are discrete representations of limited detail, even if we adopt partial ignorance, the simulated organisms will only evolve their sensory distinctions up to the granularity with which the environment is represented in the simulation program. This may be very coarse, or it may be very fine, but it will be finite. While we will be able to see our artificial organisms as making new measurements, computations, and controls over time, eventually the set of simulated possibilities will be exhausted and there will be no more emergent behavior.

In the end the functional closure of computer simulations is due to the stability properties of the physical substrates on whose motions they depend. Digital computers are *closed-state* devices because of the finite number and stable nature of the discrete attractors they utilize for their state spaces. As functional objects, they are *defined* by this discrete, stable organization. Biological organisms and mixed digital-analog devices, in contrast, as *open-state* devices can continue to proliferate new discrete attractors (new symbol primitives) because of the continuous, contingently stable nature of their energy landscapes.⁷ New distinctions can then arise out of the continuum, in a constant interplay between the differentiated and the undifferentiated, the symbolic and the indefinite.

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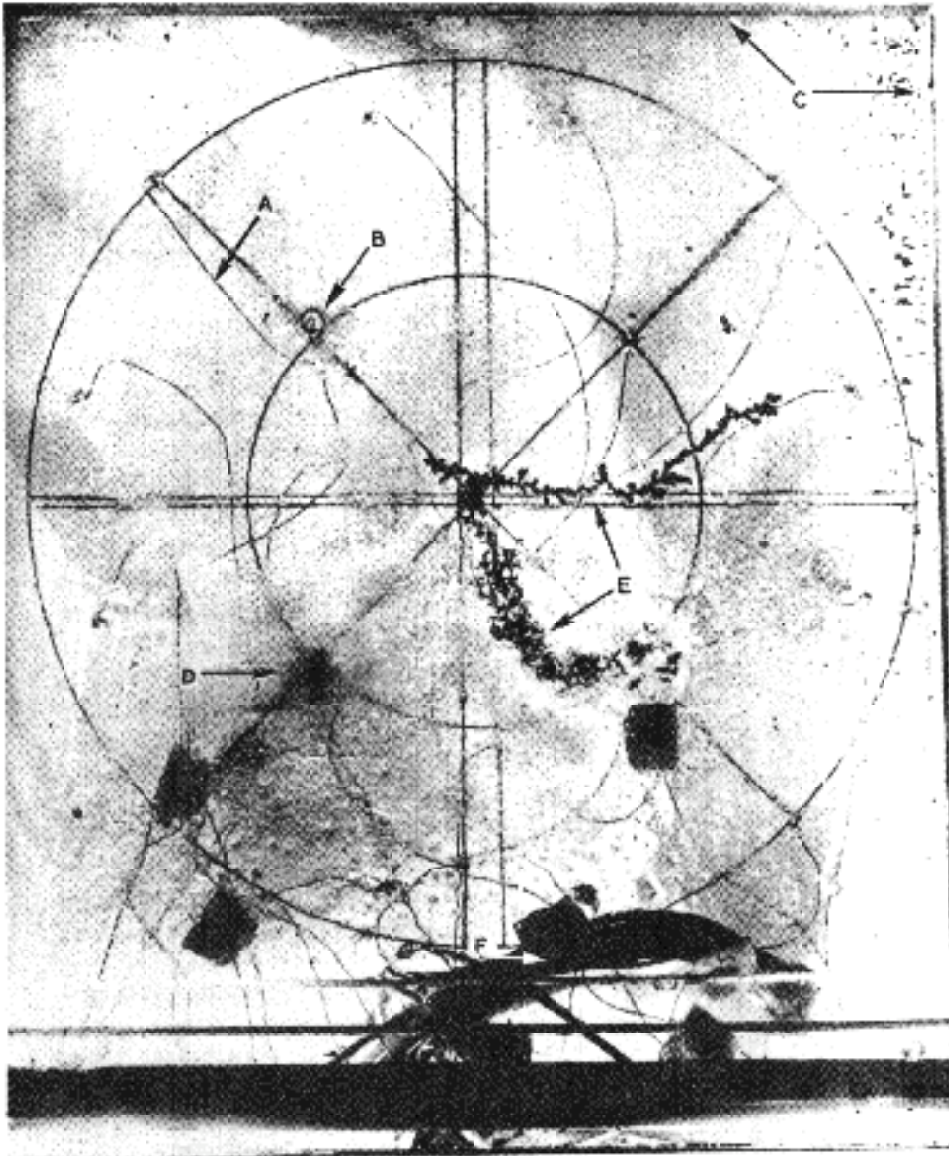


Figure 6a (Pask's device) as it should have been rendered.

Some of the footnote numbers are shifted by 1 or 2. These errors were corrected in proof but did not make it into the published version.